v1.0Baryon Deep Dive_ One-Particle Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Baryon Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Baryon Deep Dive_ One-Particle Universe V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Strangeness in LHC	Not in V1.0 scope	ALICE Run 3: strangeness enhancement in pp collisions — unexpected Lambda overproduction in high-multiplicity events	MEDIUM	New production mechanism for strange baryons in engine
Lattice QCD mass	Experimental masses	BMW 2024: Lambda mass 1115.83 MeV from first principles (exp: 1115.683 MeV). 0.13% agreement	LOW	First-principles mass calculation improves parameter accuracy
Exotic baryon states	$q\bar{q}$ mesons and qqq baryons only	LHCb 2022-2024: 3 pentaquark states + Tcc+ tetraquark confirmed. NEW particle classes required	HIGH	V1.0 data model cannot represent confirmed composite particles beyond qqq

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
WebGPU strangeness bit-packing	Conceptual only	V2.0: S□{-3,-2,-1,0,+1,+2,+3} packed in 4 bits of uint32 particle state vector	LOW	Implementation detail for performance optimization

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Exotic
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Baryon Deep Dive_ One-Particle Universe V1.0 archived | V2.0 is the active specification as of June 2026